KISSsoft Release 03/2018 F

ForEveryOne

File

Name : Shaft1
Changed by: Matin on: 26.01.2023 at: 01:12:35

Analysis of shafts, axle and beams

Input data

Coordinate system shaft: see picture W-002

Label		Shaft 1
Drawing		
Initial position (mm)		0.000
Length (mm)		280.000
Speed (1/min)		750.00
Sense of rotation: counter clockwise		
Material		C45 (1)
Young's modulus (N/mm²)		206000.000
Poisson's ratio nu		0.300
Density (kg/m³)		7830.000
Coefficient of thermal expansion	(10^-6/K)	11.500
Temperature (°C)		20.000
Weight of shaft (kg)		2.755
(Notice: Weight stands for the shaft only without considering the gears)		
Weight of shaft, including additional masses (kg)		4.604
Mass moment of inertia (kg*mm²)		1065.455
Momentum of mass GD2 (Nm²)		0.042
Position in space (°)		0.000
Gears mounted with stiffness according to ISO		
Consider deformations due to shearing		
Shear correction factor		1.100
Contact angle of rolling bearings is considered		
Tolerance field: Mean value		
Reference temperature (°C)		20.000

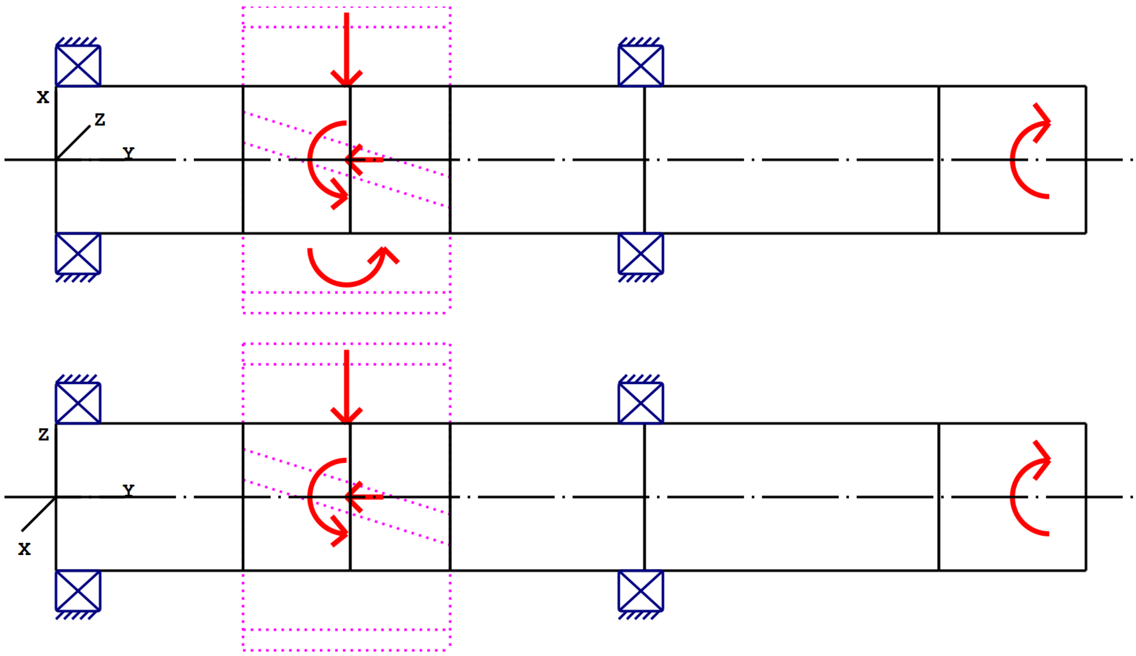


Figure: Load applications

Shaft definition (Shaft 1)

Outer contour

Cylinder (Cylinder)			0.000mm ... 80.000mm
Diameter (mm)	[d]	40.0000	
Length (mm)	[l]	80.0000	
Surface roughness (μm)	[Rz]	8.0000	

Cylinder (Cylinder)			80.000mm ... 160.000mm
Diameter (mm)	[d]	40.0000	
Length (mm)	[l]	80.0000	
Surface roughness (μm)	[Rz]	8.0000	

Cylinder (Cylinder)			160.000mm ... 240.000mm
Diameter (mm)	[d]	40.0000	
Length (mm)	[l]	80.0000	
Surface roughness (μm)	[Rz]	8.0000	

Cylinder (Cylinder)			240.000mm ... 280.000mm
Diameter (mm)	[d]	40.0000	
Length (mm)	[l]	40.0000	
Surface roughness (μm)	[Rz]	8.0000	

Forces

Type of force element	**Rope sheave**
Label in the model	Rope sheave

Position on shaft (mm)	[y_{local}]	270.0000
Position in global system (mm)	[y_{global}]	270.0000
Direction of belt force (°)		0.0000
Sheave diameter (mm)		0.0000
Length of load application (mm)		20.0000
Power (kW)		3.0000 driven (input)
Torque (Nm)		-38.1972
Axial force (N)		0.0000
Shearing force X (N)		0.0000
Shearing force Z (N)		0.0000
Bending moment X (Nm)		0.0000
Bending moment Z (Nm)		0.0000
Sum of the belt forces (N)		0.0000

Type of force element		**Cylindrical gear**
Label in the model		Cylindrical gear
Position on shaft (mm)	[y_{local}]	79.0000
Position in global system (mm)	[y_{global}]	79.0000
Operating pitch diameter (mm)		83.3030
Helix angle (°)		25.1456 right
Working pressure angle at normal section (°)		20.8359
Position of contact (°)		0.0000
Length of load application (mm)		56.3000
Power (kW)		3.0000 driving (output)
Torque (Nm)		38.1972
Axial force (N)		-430.4754
Shearing force X (N)		-385.5582
Shearing force Z (N)		-917.0663
Bending moment X (Nm)		0.0000
Bending moment Z (Nm)		-17.9299

Bearing

Label in the model		Rolling bearing
Bearing type		SKF 61908
Bearing type		Deep groove ball bearing (single row)

Bearing position (mm)	[y_{lokal}]	6.000
Bearing position (mm)	[y_{global}]	6.000
Attachment of external ring		Set fixed bearing left
Inner diameter (mm)	[d]	40.000
External diameter (mm)	[D]	62.000
Width (mm)	[b]	12.000
Corner radius (mm)	[r]	0.600
Basic static load rating (kN)	[C_0]	10.000
Basic dynamic load rating (kN)	[C]	13.800
Fatigue load rating (kN)	[C_u]	0.425
Values for approximated geometry:		
Basic dynamic load rating (kN)	[C_{theo}]	0.000
Basic static load rating (kN)	[C_{0theo}]	0.000
Correction factor Basic dynamic load rating	[f_C]	1.000
Correction factor Basic static load rating	[f_{C0}]	1.000

Label in the model		Rolling bearing
Bearing type		SKF 61908
Bearing type		Deep groove ball bearing (single row)

Bearing position (mm)	[y_{lokal}]	159.000
Bearing position (mm)	[y_{global}]	159.000
Attachment of external ring		Set fixed bearing right
Inner diameter (mm)	[d]	40.000
External diameter (mm)	[D]	62.000
Width (mm)	[b]	12.000
Corner radius (mm)	[r]	0.600
Basic static load rating (kN)	[C_0]	10.000
Basic dynamic load rating (kN)	[C]	13.800
Fatigue load rating (kN)	[C_u]	0.425
Values for approximated geometry:		
Basic dynamic load rating (kN)	[C_{theo}]	0.000
Basic static load rating (kN)	[C_{0theo}]	0.000
Correction factor Basic dynamic load rating		
	[f_C]	1.000
Correction factor Basic static load rating		
	[f_{C0}]	1.000

Shaft 'Shaft 1': Cylindrical gear 'Cylindrical gear' (y= 65.4250 (mm)) is taken into account as component of the shaft.
EI (y= 50.8500 (mm)): 25886.7235 (Nm²), EI (y= 80.0000 (mm)): 25886.7235 (Nm²), m (yS= 65.4250 (mm)): 0.9572 (kg)
Jp: 0.0003 (kg*m²), Jxx: 0.0002 (kg*m²), Jzz: 0.0002 (kg*m²)

Shaft 'Shaft 1': Cylindrical gear 'Cylindrical gear' (y= 93.5750 (mm)) is taken into account as component of the shaft.
EI (y= 80.0000 (mm)): 25886.7235 (Nm²), EI (y= 107.1500 (mm)): 25886.7235 (Nm²), m (yS= 93.5750 (mm)): 0.8915 (kg)
Jp: 0.0002 (kg*m²), Jxx: 0.0001 (kg*m²), Jzz: 0.0001 (kg*m²)

Results

Shaft

Maximum deflection (µm)	7.696
Position of the maximum (mm)	80.000
Mass center of gravity (mm)	140.000
Total axial load (N)	-430.475
Torsion under torque (°)	-0.018

Bearing

Probability of failure	[n]	10.00	%
Axial clearance	[u_A]	10.00	µm
Lubricant	Oil: ISO-VG 220		
Lubricant - service temperature	[T_B]	20.00	°C

Rolling bearings, classical calculation (contact angle considered)

Shaft 'Shaft 1' Rolling bearing 'Rolling bearing'

Position (Y-coordinate)	[y]	6.00	mm
Dynamic equivalent load	[P]	1.06	kN
Equivalent load	[P_0]	0.58	kN
Life modification factor for reliability[a_1]		1.000	
Basic bearing rating life	[L_{nh}]	48553.54	h
Operating viscosity	[ν]	912.87	mm²/s
Static safety factor	[S_0]	17.19	
Bearing reaction force	[Fx]	0.319	kN
Bearing reaction force	[Fy]	0.430	kN
Bearing reaction force	[Fz]	0.487	kN
Bearing reaction force	[Fr]	0.582	kN (56.77°)
Oil level	[H]	0.000	mm
Rolling moment of friction	[M_{rr}]	0.143	Nm
Sliding moment of friction	[M_{sl}]	0.009	Nm
Moment of friction, seals	[M_{seal}]	0.000	Nm
Moment of friction for seals determined according to SKF main catalog 17000/1 EN:2018			
Moment of friction flow losses	[M_{drag}]	0.000	Nm
Torque of friction	[M_{loss}]	0.152	Nm
Power loss	[P_{loss}]	11.912	W
The moment of friction is calculated according to the details in SKF Catalog 2018.			
The calculation is always performed with a coefficient for additives in the lubricant µbl=0.15.			
Displacement of bearing	[u_x]	-3.601	µm
Displacement of bearing	[u_y]	-10.000	µm
Displacement of bearing	[u_z]	-5.411	µm
Displacement of bearing	[u_r]	6.500	µm (-123.64°)
Misalignment of bearing	[r_x]	-0.032	mrad (-0.11')
Misalignment of bearing	[r_y]	-0.000	mrad (0')
Misalignment of bearing	[r_z]	-0.006	mrad (-0.02')
Misalignment of bearing	[r_r]	0.033	mrad (0.11')

Shaft 'Shaft 1' Rolling bearing 'Rolling bearing'

Position (Y-coordinate)	[y]	159.00	mm
Dynamic equivalent load	[P]	0.47	kN
Equivalent load	[P_0]	0.47	kN
Life modification factor for reliability[a_1]		1.000	
Basic bearing rating life	[L_{nh}]	564065.57	h
Operating viscosity	[ν]	912.87	mm²/s
Static safety factor	[S_0]	21.30	
Bearing reaction force	[Fx]	0.067	kN
Bearing reaction force	[Fy]	0.000	kN
Bearing reaction force	[Fz]	0.465	kN
Bearing reaction force	[Fr]	0.470	kN (81.83°)
Oil level	[H]	0.000	mm
Rolling moment of friction	[M_{rr}]	0.044	Nm
Sliding moment of friction	[M_{sl}]	0.002	Nm
Moment of friction, seals	[M_{seal}]	0.000	Nm
Moment of friction for seals determined according to SKF main catalog 17000/1 EN:2018			
Moment of friction flow losses	[M_{drag}]	0.000	Nm
Torque of friction	[M_{loss}]	0.046	Nm
Power loss	[P_{loss}]	3.611	W
The moment of friction is calculated according to the details in SKF Catalog 2018.			
The calculation is always performed with a coefficient for additives in the lubricant µbl=0.15.			

Displacement of bearing	[u_x]	-0.916	µm
Displacement of bearing	[u_y]	-10.094	µm
Displacement of bearing	[u_z]	-6.435	µm
Displacement of bearing	[u_r]	6.500	µm (-98.1°)
Misalignment of bearing	[r_x]	0.019	mrad (0.06')
Misalignment of bearing	[r_y]	-0.109	mrad (-0.37')
Misalignment of bearing	[r_z]	-0.025	mrad (-0.09')
Misalignment of bearing	[r_r]	0.032	mrad (0.11')

Damage (%) [Lreq] (20000.000)

Bin no	B1	B2
1	41.19	3.55
Σ	41.19	3.55

Utilization (%) [Lreq] (20000.000)

B1	B2
74.41	32.85

Note: Utilization = (Lreq/Lh)^(1/k)
Ball bearing: k = 3, roller bearing: k = 10/3

B1: Rolling bearing
B2: Rolling bearing

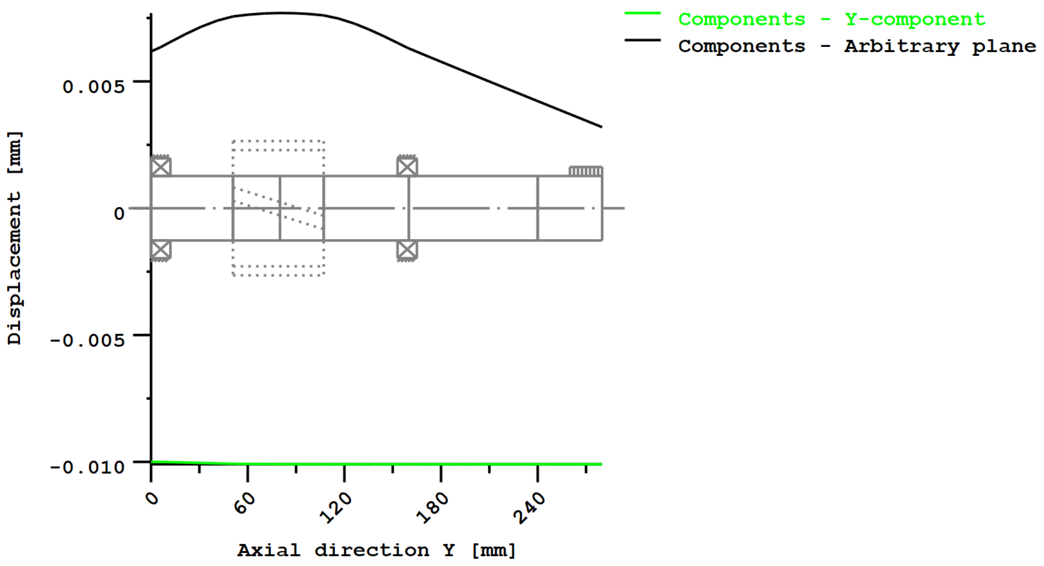
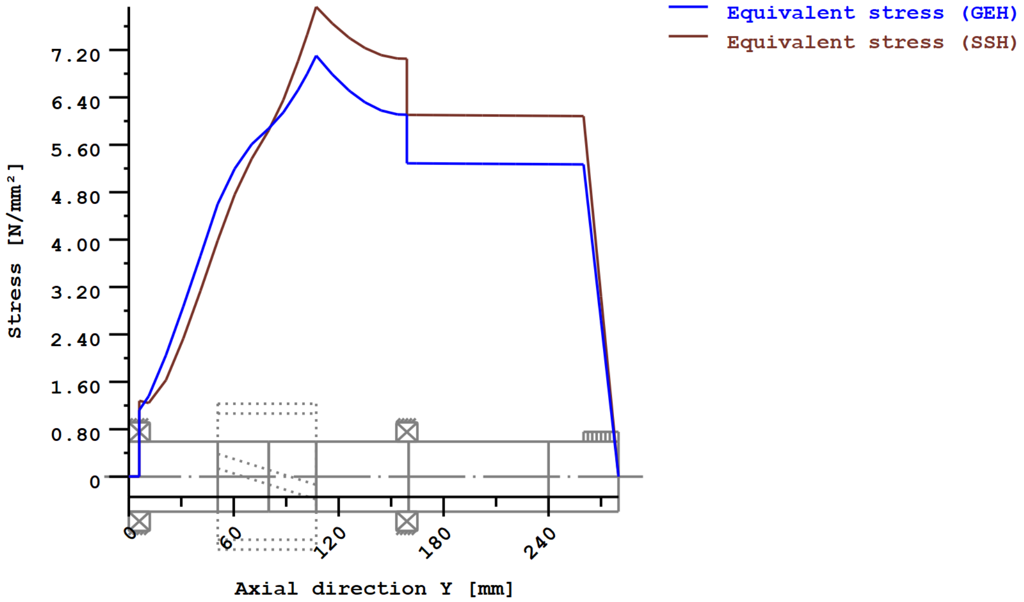


Figure: Deformation (bending etc.) (Arbitrary plane 248.7155377 121)



Nominal stresses, without taking into account stress concentrations
GEH(von Mises): sigV = ((sigB+sigZ,D)^2 + 3*(tauT+tauS)^2)^1/2
SSH(Tresca): sigV = ((sigB-sigZ,D)^2 + 4*(tauT+tauS)^2)^1/2

Figure: Equivalent stress

Strength calculation according to DIN 743:2012

Summary

Shaft 1

Material	C45 (1)
Material type	Through hardened steel
Material treatment	unalloyed, through hardened
Surface treatment	No

Calculation of endurance limit and the static strength

Calculation for load case 2 (σav/σmv = const)

Cross section	Position (Y-Coord) (mm)	
A-A	96.77	Smooth shaft

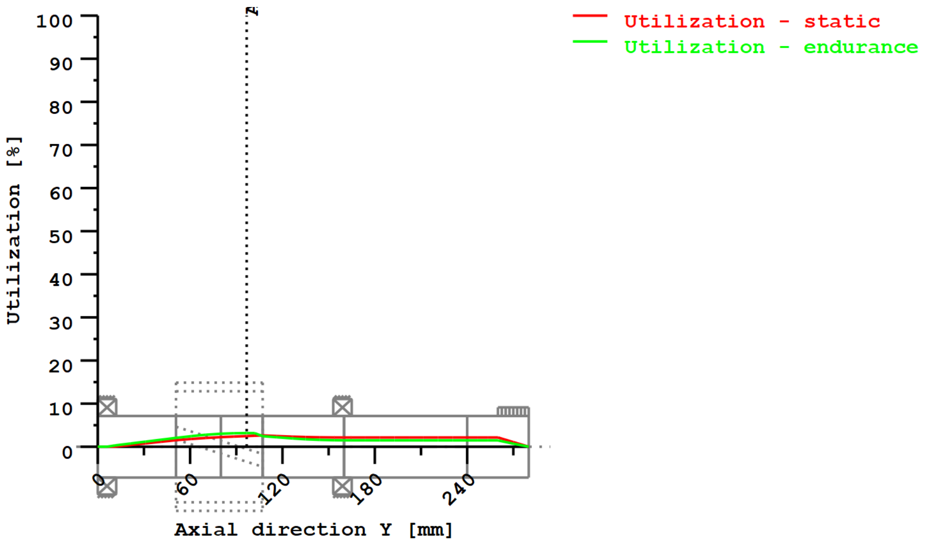
Results:

Cross section	Kfb	Kfσ	K2d	SD	SS
A-A	1.00	0.90	0.89	37.88	48.37
Required safeties:				1.20	1.20

Abbreviations:
Kfb: Notch factor bending
Kfσ: Surface factor
K2d: size factor bending
SD: Safety endurance limit
SS: Safety against yield point

Utilization (%) [Smin/S]

Cross section	Static	Endurance
A-A	2.481	3.168
Maximum utilization (%)	[A]	3.168



Utilization = Smin/S (%)

Figure: Strength

Calculation details

General statements

Label	Shaft 1		
Drawing			
Length (mm)		[l]	280.00
Speed (1/min)		[n]	750.00

Material	C45 (1)
Material type	Through hardened steel
Material treatment	unalloyed, through hardened
Surface treatment	No

	Tension/Compression	Bending	Torsion	Shearing
Load factor static calculation	1.700	1.700	1.700	1.700
Load factor endurance limit	1.000	1.000	1.000	1.000

Reference diameter material (mm)	[dB]	16.00
σB according to DIN 743 (at dB) (N/mm²)	[σB]	700.00
σS according to DIN 743 (at dB) (N/mm²)	[σS]	490.00
[σzdW] (bei dB) (N/mm²)		280.00
[σbW] (bei dB) (N/mm²)		350.00
[τtW] (bei dB) (N/mm²)		210.00
Thickness of raw material (mm)	[dWerkst]	45.00

Material data calculated according DIN743/3 with K1(d)
Material strength calculated from size of raw material
Geometric size factor K1d calculated from raw material diameter

[σBeff] (N/mm^2)	618.27
[σSeff] (N/mm^2)	415.18
[σbF] (N/mm^2)	498.22
[τtF] (N/mm^2)	287.65
[σBRand] (N/mm^2)	628.00
[σzdW] (N/mm^2)	247.31
[σbW] (N/mm^2)	309.13
[τtW] (N/mm^2)	185.48

Endurance limit for single stage use

Calculation for load case 2 (σ.av/σ.mv = const)

Cross section 'A-A' Smooth shaft

Comment		
Position (Y-Coordinate) (mm)	[y]	96.767
External diameter (mm)	[da]	40.000
Inner diameter (mm)	[di]	0.000
Notch effect		Smooth shaft
Mean roughness (µm)	[Rz]	8.000

	Tension/Compression	Bending	Torsion	Shearing
Load: (N) (Nm)				
Mean value [Fzdm, Mbm, Tm, Fqm]	-39.7	0.0	15.6	0.0
Amplitude [Fzda, Mba, Ta, Fqa]	39.7	27.3	15.6	276.6
Maximum value [Fzdmax, Mbmax, Tmax, Fqmax]	-135.0	46.5	53.0	470.3
Cross section, moment of resistance: (mm^2)				
[A, Wb, Wt, A]	1256.6	6283.2	12566.4	1256.6
Stresses: (N/mm^2)				
[σzdm, σbm, τm, τqm] (N/mm^2)	-0.032	0.000	1.240	0.000
[σzda, σba, τa, τqa] (N/mm^2)	0.032	4.351	1.240	0.294
[σzdmax,σbmax,τmax,τqmax] (N/mm^2)	-0.107	7.397	4.214	0.499

Technological size influence	[K1(σB)]	0.883
	[K1(σS)]	0.847

		Tension/Compression	Bending	Torsion
Notch effect coefficient	[ß]	1.000	1.000	1.000
Geometrical size influence	[K2(d)]	1.000	0.888	0.888
Influence coefficient surface roughness	[KF]	0.903	0.903	0.944
Surface stabilization factor	[KV]	1.000	1.000	1.000
Total influence coefficient	[K]	1.108	1.234	1.185

Present safety for endurance limit:

Equivalent mean stress (N/mm^2)	[σmV]	2.147
Equivalent mean stress (N/mm^2)	[τmV]	1.239

Fatigue limit of part (N/mm^2)	[σWK]	223.223	250.572	156.505

Influence coefficient of mean stress sensitivity.

	[ψσK]	0.220	0.254	0.145
Permissible amplitude (N/mm²)	[σADK]	6.021	222.657	136.698
Safety against fatigue	[S]		37.884	
Required safety against fatigue	[Smin]		1.200	
Result (%)	[S/Smin]		3157.0	

Present safety
for proof against exceed of yield point:

Static notch sensitivity factor	[K2F]	1.000	1.200	1.200
Increase coefficient	[γF]	1.000	1.000	1.000
Yield stress of part (N/mm²)	[σFK]	415.181	498.217	287.646
Safety yield stress	[S]		48.365	
Required safety	[Smin]		1.200	
Result (%)	[S/Smin]		4030.4	

Remarks:
- The shearing force is not considered in the analysis specified in DIN 743.
- Cross section with interference fit:
  The notching factor for the light fit case is no longer defined in DIN 743.
  The values are imported from the FKM-Guideline..

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